

Nom Prénom : GOOGLE GEMINI

Quick Test 1 : 9/10/2026

0.1 Exo 1 : Mettre les expressions suivantes sous forme algébrique

1. $\frac{2+3i}{1+i}$

$$\begin{aligned}\frac{2+3i}{1+i} &= \frac{(2+3i)(1-i)}{1^2+1^2} \\ &= \frac{2-2i+3i+3}{2} \\ &= \frac{5+i}{2} = \frac{5}{2} + \frac{1}{2}i\end{aligned}$$

2. $(3-i)^2 + \operatorname{Re} \overline{1-2i}$

$$\begin{aligned}(3-i)^2 + \operatorname{Re}(\overline{1-2i}) &= (9-6i-1) + \operatorname{Re}(1+2i) \\ &= 8-6i+1 \\ &= 9-6i\end{aligned}$$

3. $3 \times \left(\frac{2+3i}{5-i} \right)$

$$\begin{aligned}3 \times \left(\frac{2+3i}{5-i} \right) &= 3 \times \frac{(2+3i)(5-i)}{5^2+1^2} \\ &= 3 \times \frac{10-2i+15i+3}{25+1} \\ &= 3 \times \frac{13+13i}{26} = 3 \times \frac{1+i}{2} = \frac{3}{2} + \frac{3}{2}i\end{aligned}$$

4. $\frac{1-2ai}{2a+i}$ pour $a \in \mathbb{R}$

$$\begin{aligned}\frac{1-2ai}{2a+i} &= \frac{(1-2ai)(2a-i)}{(2a)^2+1^2} \\ &= \frac{2a-i-4a^2i-2a}{4a^2+1} \\ &= \frac{(-1-4a^2)i}{4a^2+1} = -i\end{aligned}$$

0.2 Exo 2 : Mettre les expressions suivantes sous forme exponentielle

1. $Z = (1+i\sqrt{3})^3$

$$\begin{aligned}1+i\sqrt{3} &= 2e^{i\pi/3} \\ Z &= (2e^{i\pi/3})^3 \\ &= 8e^{i\pi} = -8 = 8e^{i\pi}\end{aligned}$$

2. $Z = \frac{\sqrt{3}-i}{1+i}$

$$\begin{aligned}\sqrt{3}-i &= 2e^{-i\pi/6} \quad \text{et} \quad 1+i = \sqrt{2}e^{i\pi/4} \\ Z &= \frac{2e^{-i\pi/6}}{\sqrt{2}e^{i\pi/4}} = \sqrt{2}e^{i(-\pi/6-\pi/4)} \\ &= \sqrt{2}e^{-i\frac{5\pi}{12}}\end{aligned}$$

3. $Z = \frac{\exp(i\frac{\pi}{4}) \times \exp(-i\frac{\pi}{2})}{\exp(i\frac{\pi}{6})}$

$$\begin{aligned} Z &= \exp\left(i\left(\frac{\pi}{4} - \frac{\pi}{2} - \frac{\pi}{6}\right)\right) \\ &= \exp\left(i\left(-\frac{5\pi}{12}\right)\right) \end{aligned}$$

$$4. \quad Z = \frac{(\exp(i\frac{\pi}{6}))^4}{(\exp(-i\frac{\pi}{4}))^2}$$

$$\begin{aligned} Z &= \frac{\exp(i\frac{2\pi}{3})}{\exp(-i\frac{\pi}{2})} \\ &= \exp\left(i\left(\frac{2\pi}{3} - \left(-\frac{\pi}{2}\right)\right)\right) = \exp\left(i\frac{7\pi}{6}\right) \end{aligned}$$

0.3 Exo 3 : Résoudre les équations :

$$1. \quad X^2 - 4X + 13 = 0$$

$$\begin{aligned} \Delta &= (-4)^2 - 4(1)(13) = 16 - 52 = -36 = (6i)^2 \\ X &= \frac{4 \pm 6i}{2} = 2 \pm 3i \\ S &= \{2 + 3i, 2 - 3i\} \end{aligned}$$

$$2. \quad \text{Résoudre à la fois : } X^2 = 8 - 6i \quad \text{et} \quad X^2 = 8 + 6i$$

Pour $X^2 = 8 - 6i$, posons $X = x + iy$:

$$\begin{cases} x^2 - y^2 = 8 \\ 2xy = -6 \implies xy = -3 \\ x^2 + y^2 = \sqrt{8^2 + (-6)^2} = 10 \end{cases}$$

En combinant $x^2 - y^2 = 8$ et $x^2 + y^2 = 10$, on obtient $2x^2 = 18 \implies x = \pm 3$, d'où les solutions $3 - i$ et $-3 + i$.
 Pour $X^2 = 8 + 6i$, les solutions sont les conjuguées : $3 + i$ et $-3 - i$.

3. $X^2 - (3 + i)X + (2 + 2i) = 0$

$$\begin{aligned}\Delta &= [-(3 + i)]^2 - 4(1)(2 + 2i) \\ &= (9 + 6i - 1) - (8 + 8i) \\ &= (8 + 6i) - (8 + 8i) = -2i\end{aligned}$$

Cherchons une racine carrée $\delta = x + iy$ de $-2i$ telle que $\delta^2 = -2i$:

$$\begin{cases} x^2 - y^2 = 0 \\ 2xy = -2 \implies xy = -1 \\ x^2 + y^2 = |-2i| = 2 \end{cases} \implies 2x^2 = 2 \implies x = \pm 1, y = \mp 1$$

On choisit $\delta = 1 - i$. Les solutions sont :

$$\begin{aligned}X &= \frac{(3 + i) \pm (1 - i)}{2} \\ X_1 &= \frac{4}{2} = 2, \quad X_2 = \frac{2 + 2i}{2} = 1 + i \\ S &= \{2, 1 + i\}\end{aligned}$$